

19.07.2026

Lecture - 1

Discrete Signals & Systems

Signals: A signal is a measurable physical quantity (such as voltage, pressure) that varies over an independent variable (like time or space) to carry meaningful information.

Physically: It is an observable parameters carrying message or data (e.g. sound waves, heartbeat traces).

Mathematically: It is expressed as a function of one of or more independent variables written as:

$x(t)$ $\longrightarrow$ continuous time.

$x[n]$ $\longrightarrow$ discrete

System: ~~A signal is a measurable physical quantity~~

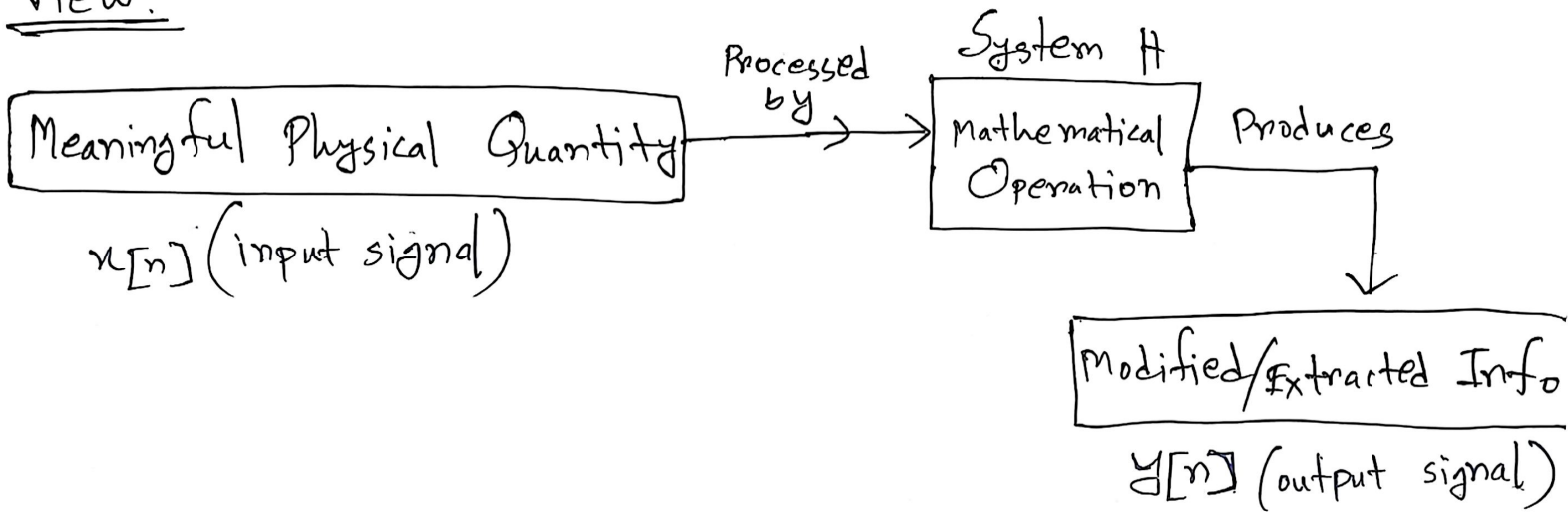
A system is set of interconnected components or an algorithm that process an input signal to extract, modify or transform its information to yield a specific output signal.

Physically: It is the entity or device doing the work (e.g. an amplifier, a noise-canceling circuit).

Mathematically: It represents a transform operator ^②
 H that maps an input sequence $x[n]$ into
 an output sequence $y[n]$.

$$y[n] = H\{x[n]\}$$

View:



1.3

Continuous and Discrete Time Signal

3 Types: ① Continuous Time Signal

② Discrete Time Signals.

③ Digital Signal

Audio Signal $\rightarrow$ time $\rightarrow A(t)$
(A)

Image (I) $\rightarrow$ space $\rightarrow I(s)$

Video $\rightarrow$ time, space $\rightarrow V(s, t)$

① Continuous - Time (CT) Signal: A continuous-time signal is a physical or mathematical signal that is defined

continuously across a continuous range of time. ☺

→ Mathematical Representation. It is denoted as $x(t)$, where the independent signal variable t is a continuous variable. Here $(t \in \mathbb{R})$.

At
→ Key Characteristic: Any exact instant of time t , no matter how ~~infinite~~ immeasurably small, $x(t)$ has defined value. ~~is~~ for representation inside parenthesis $()$ as $x(t)$, $y(t)$, $u(t)$.

Examples: Human voice in air, body temperature throughout a day.

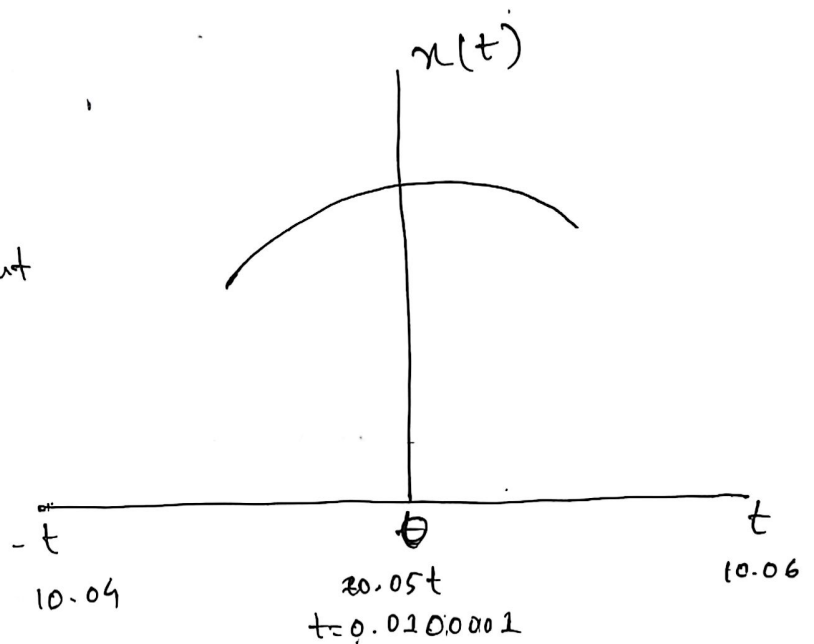


Fig. continuous-time signal

② Discrete-Time (DT) Signal: A Discrete-Time Signal is a signal that is defined only at discrete points in time. It is essentially a sequence of numbers indexed by an integer.

(4)

⇒ Mathematical Representation: Independent variable n , putting inside square parenthesis $[n]$. Denoted as $x[n]$, $y[n]$, $u[n]$ etc. n is integer index ($n \in \mathbb{Z}$) representing discrete time steps.

⇒ Key characteristics: $x[n]$ is undefined for non-integer values of n . For example $x[2.5]$ does not exist.

Examples: End-of-the-day stock prices, monthly sales figures.

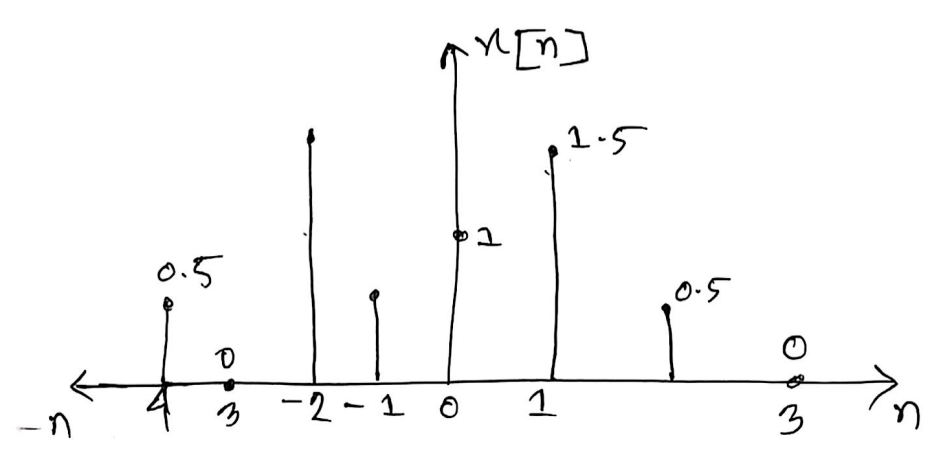
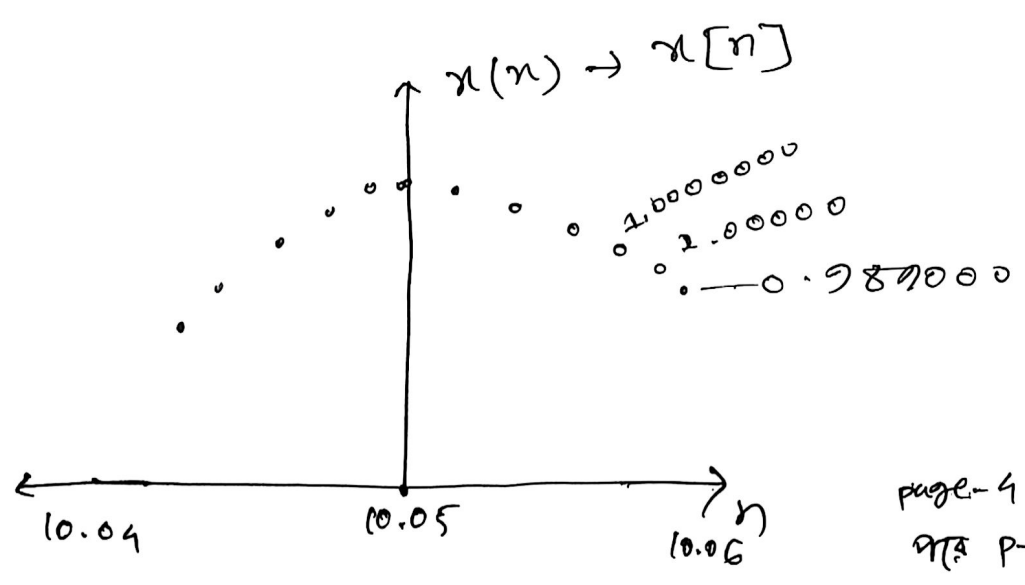


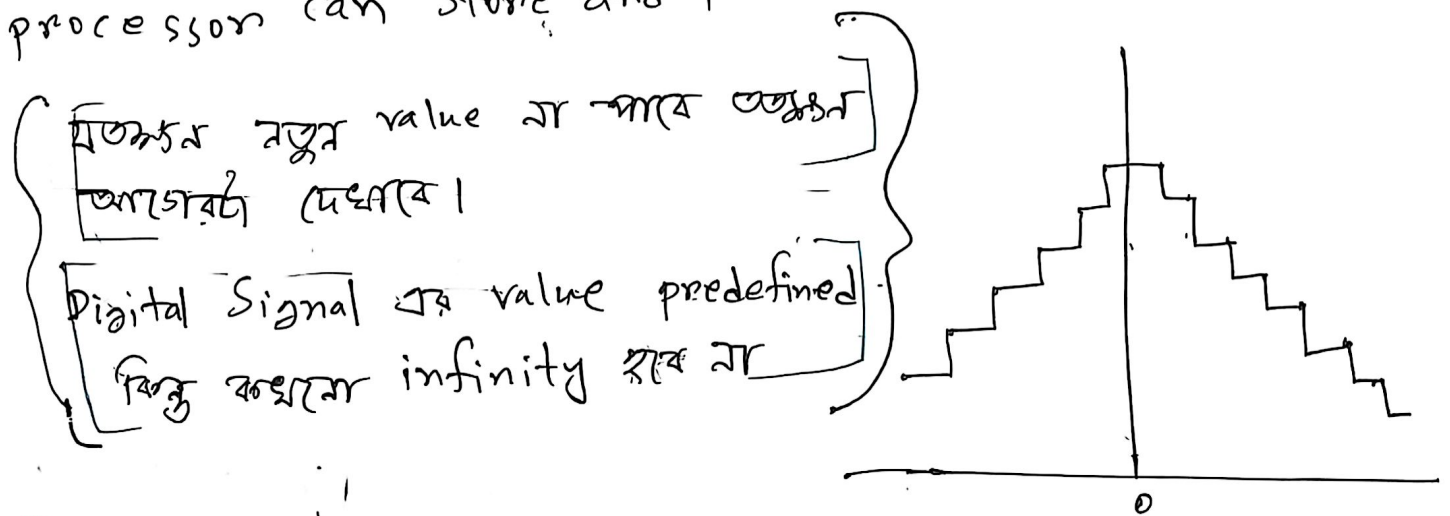
Fig: Discrete-Time Signal



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③ Digital Signal: A Digital Signal is a special type of ⁽⁵⁾ discrete-time signal where both time and amplitude are discrete (quantized)

More precisely, a Digital Signal is a discrete-time signal whose values have been quantized so that a digital computer or processor can store and process it as binary ~~number~~ numbers.



Characteristics:

① Sampling: The signal is evaluated only at specific integer intervals ($n=0, 1, 2, \dots$).

② Quantization: The value/magnitude of the signal at each sample is rounded to the nearest level from a finite set of allowed values.

③ Representation: ~~is~~ Usually represented in binary code (bits) inside computers, DSP chips and ^{D.C. systems} _{digital comm.}

Discrete Signal Representation

1. Formula $\rightarrow x[n] = \begin{cases} (\frac{1}{a})^n, & n \geq 0 \\ 0, & n < 0 \end{cases}$

2. Sequence $\rightarrow x[n] = \left\{ 0, 0, 0, \underset{\substack{\uparrow \\ n=0}}{1}, \frac{1}{a}, \frac{1}{a^2}, \frac{1}{a^3}, \dots \right\}$

arrow থাকলে অর্থাৎ বসাবার $n=0$ শুরু হবে

3. Graphical Representation: $= \left\{ 1, \frac{1}{a}, \frac{1}{a^2}, \frac{1}{a^3} \right\}$

Note: (যদি 12) sequence-এ কোনো অসম্পূর্ণ value বা ফাঁকা থাকে থাকলে তখন সেটিকে 0 বসিয়ে দিতে হবে।
 $\rightarrow$ এই ঘটনাকে Zero padding বলে।

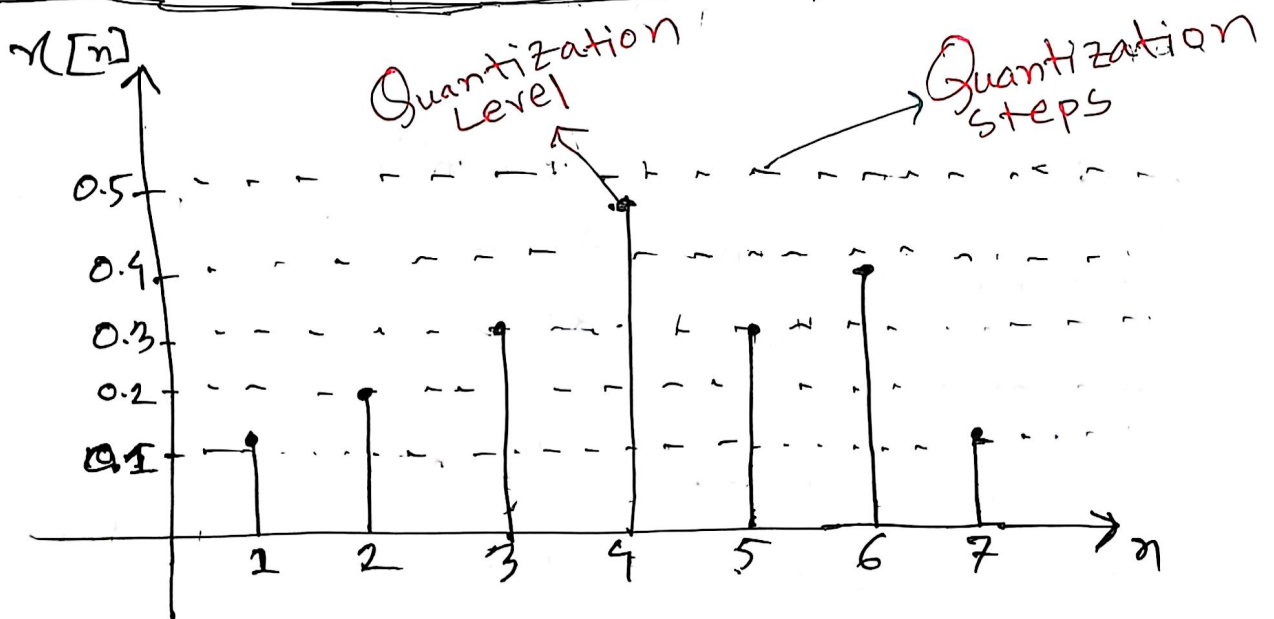


Fig: Digital Signal

Example: Graphically represent $x[n] = \{3, 2, 5, 4, 6, 8, 2\}$

Soln: Here, $x[n] = \{3, 2, 5, 4, 6, 8, 2\}$

$$= \left\{ \underset{n=0}{3}, \underset{n=1}{2}, \overset{n=2}{5}, \underset{n=3}{4}, \overset{n=4}{6}, \underset{n=5}{8}, \underset{n=6}{2} \right\}$$

$$\therefore n = 0, 1, 2, 3, 4$$

$$x[0] = 5$$

$$x[1] = 4$$

$$x[2] = 6$$

$$x[3] = 8$$

$$x[4] = 2$$

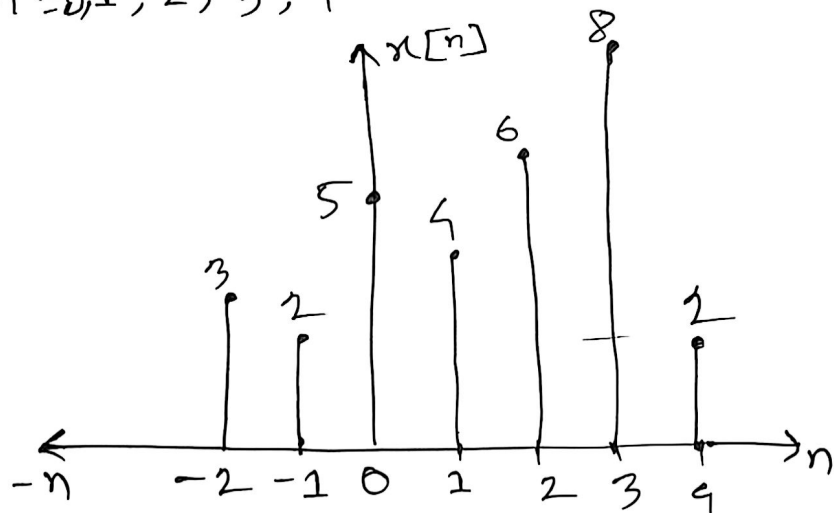


Fig: Graphical representation of $x[n]$

Example: 1.1 Graphically represent $x[n] = \{1, 0, -1, 1\}$

Here, $x[n] = \left\{ \underset{n=0}{1}, \underset{n=1}{0}, \underset{n=2}{-1}, \underset{n=3}{1} \right\}$

$$\therefore n = 0, 1, 2, 3$$

$$x[0] = 1$$

$$x[1] = 0$$

$$x[2] = -1$$

$$x[3] = 1$$

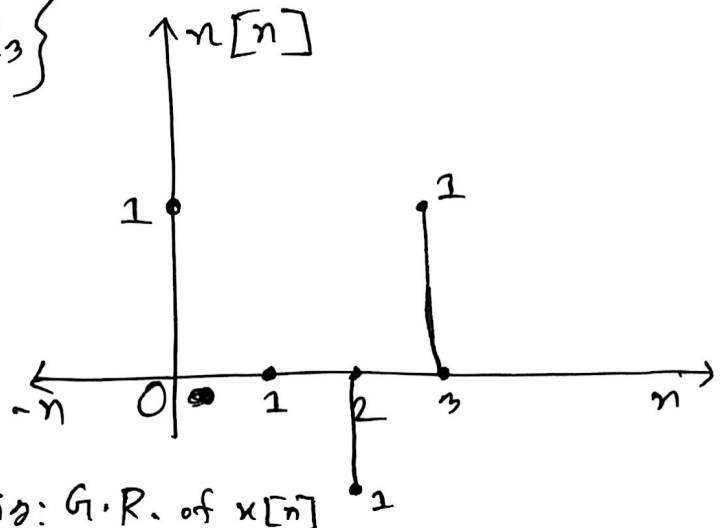
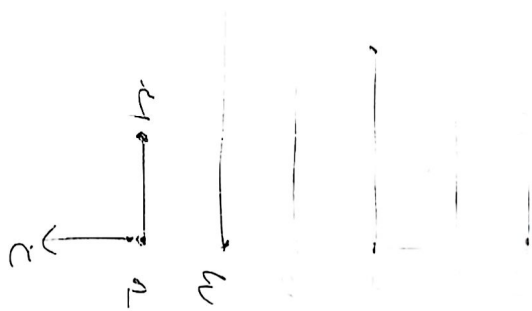


Fig: G.R. of $x[n]$

Example - 1.2 Graphically represent the following sequence.

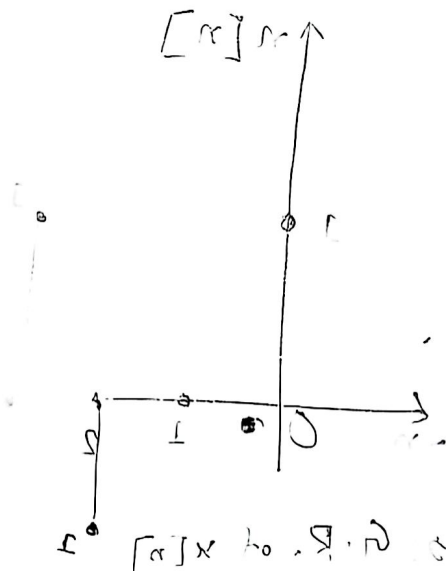
$$x[n] = \{-2, 1, 0, 1, 2, 0, 1\}$$

↑



$x[n]$ is a discrete-time signal.

$$\{-2, 1, 0, 1\} = [x]$$



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~~Digit~~

1.4 Basic Discrete Signal

1. Unit Impulse Signal

CT-Signal vs DT-Signal

Feature	Continuous-Time Signal	Discrete-Time Signal
Notation	$x(t)$ → uses round parenthesis	$x[n]$ → uses square bracket
Independent Variable	t → Continuous; real number	n → Discrete, integer sequence
Domain	Defined at all time instants	Defined only at integer indices
Representation	Smooth continuous curves/waveforms	Sequence of discrete pulses/stems
Typical Context	Analog Circuits, Physical World	Computers, DSP microprocessors, digital algorithms

Key Distinction (Diff) (DT-Signal vs D.S.)

DT-Signal $x[n]$: Discrete in time, but amplitude can take any continuous real value (unquantized).

D.S: Discrete in time AND discrete in amplitude
 that means quantized into a fixed set of levels,
 such as binary 0s and 1s.

Summary

<u>Signal Type</u>	<u>Time Domain</u>	<u>Amplitude Domain</u>	<u>Examples</u>
<u>Analog (CT)</u> Analog	Continuous (t)	Continuous	Raw voice from a mic.
Discrete-Time (DT)	Discrete (n)	Continuous	Unquantized sampled voltage
Digital	Discrete (n)	Discrete (quantized)	MP3 audio file, JPEG image bits

1.4

Basic Discrete Signal

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1.4.1

The Unit Impulse Sequence ($\delta[n]$): Also known as the Kronecker delta function, the unit impulse sequence is the most fundamental building block in discrete-time signal processing.

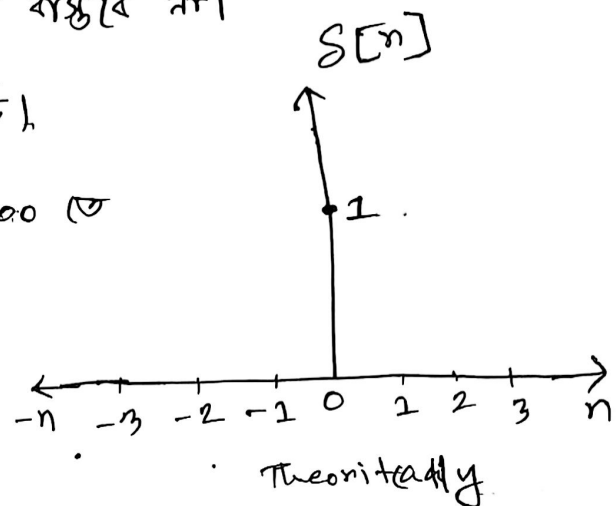
Impulse হল একক স্ক্যাল
 Impulse $\rightarrow$ Single pulse - 1
 1 at $n=0$

~~0~~ time এ ১ থাকার কিন্তু এটা বাক্য নহ।
(zero) duration

কারণ যদি ১ দৈর্ঘ্য ০ থাকে। অর্থাৎ: যদি।

50 second এর পরেই ০১ প্রকৃতি দেখানোর ০০ ৩

সিহ্ন থাকে অর্থাৎ ০১.



Mathematically:

$$S[n] = \begin{cases} 1, & n=0 \\ 0, & n \neq 0 \end{cases}$$

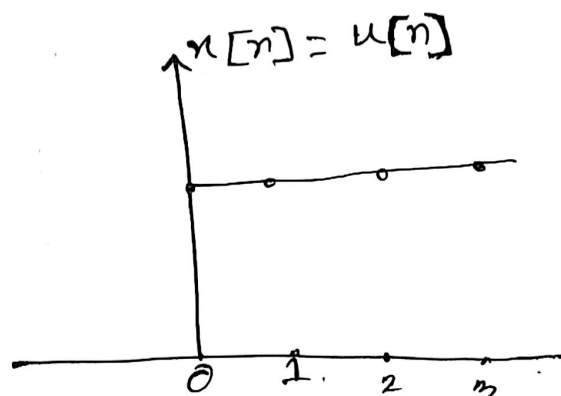
In short, the Unit Impulse Sequence denoted as $S[n]$, is a discrete signal that has a value of 1 at $n=0$ and 0 everywhere else.

1.4.2 The Unit Step Sequence: $\rightarrow (u[n])$ A discrete signal that turns ON (value = 1) at $n=0$ and stays 1 for all positive values of n .

Usage: Models a digital switch turning on and staying on permanently

Mathematically:

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$



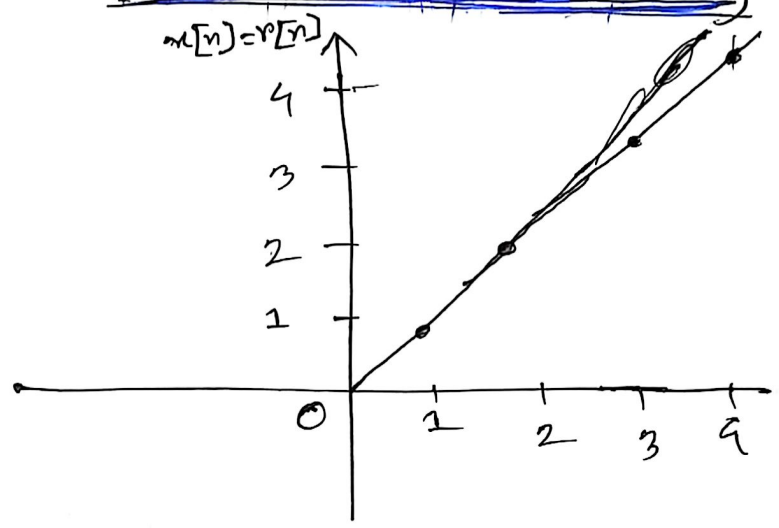
1.4.3 The Unit Ramp Signal ($r[n]$): The Unit Ramp Sequence is signal whose amplitude grows linearly with discrete time n , starting at $n=0$.

value grows linearly with sample index n

Mathematically:

$$r[n] = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

or $r[n] = n \cdot u[n]$

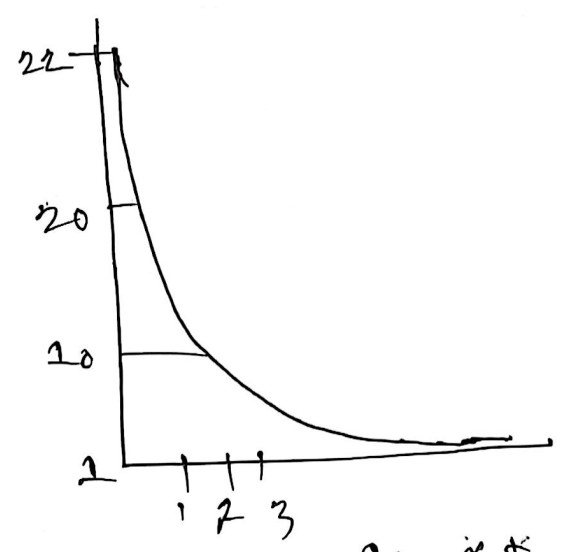


key idea: Values step up as n ; 0, 1, 2, 3, 4, ... for $n \geq 0$

1.4.6 Exponential Signal (a^n): $Z = Ae^{\alpha t}$ represents a

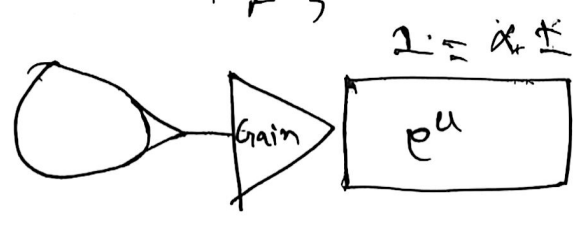
Continuous-Time Real Exponential Signal.

$Z = Ae^{\alpha t}$ $\longleftrightarrow$ $\alpha \pm$ (Gain)
 $\downarrow$ $t \rightarrow$ time
 change $A \rightarrow$ Amplitude
 Z



Convert to D-T form:

$$r[n] = Z(nT_s) = Ae^{\alpha(nT_s)}$$



$$= A(e^{aT_s})^n$$

$$= A a^n \quad \left[\text{Here: } a = e^{aT_s} \right]$$

A sequence defined by $x[n] = a^n$, where its shape depends entirely on the base value a .

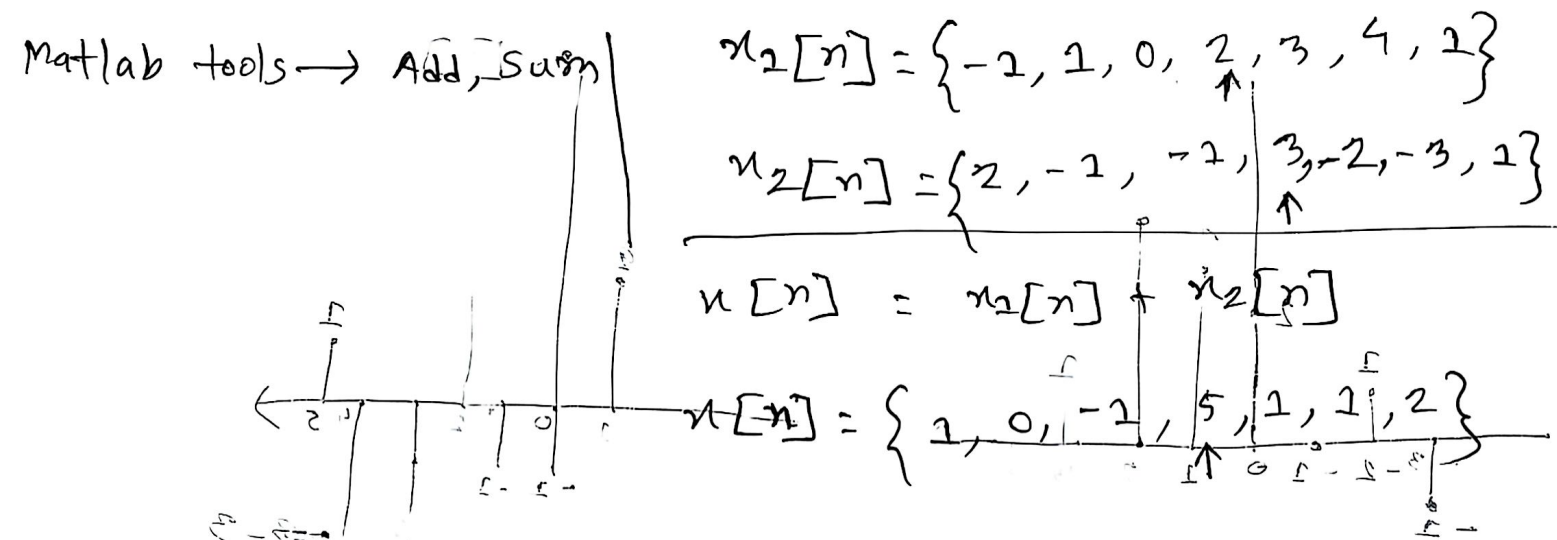
Note: → practical signal ~~cannot~~ infinity ~~or~~ ~~or~~ zero

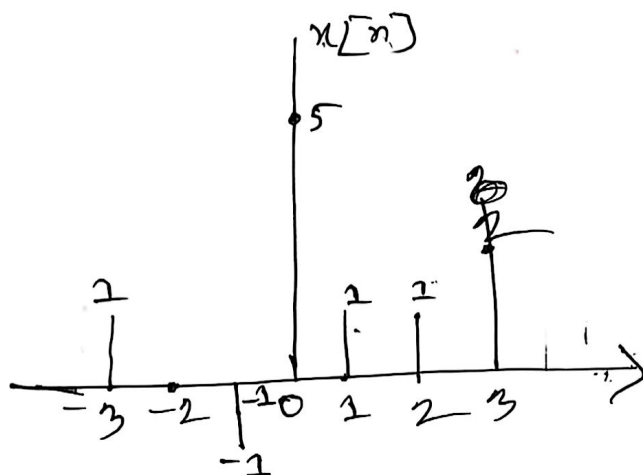
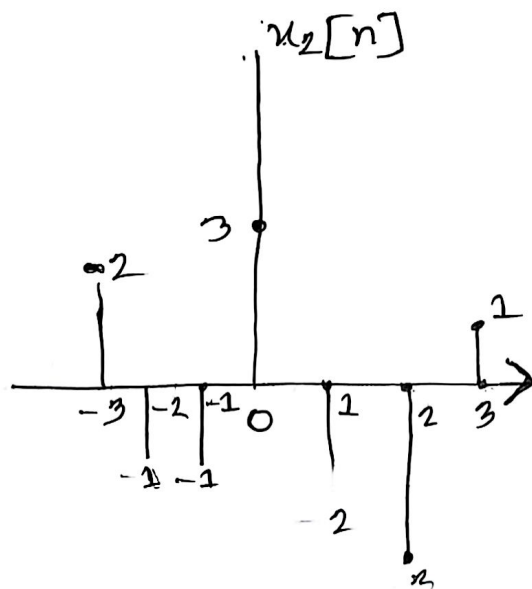
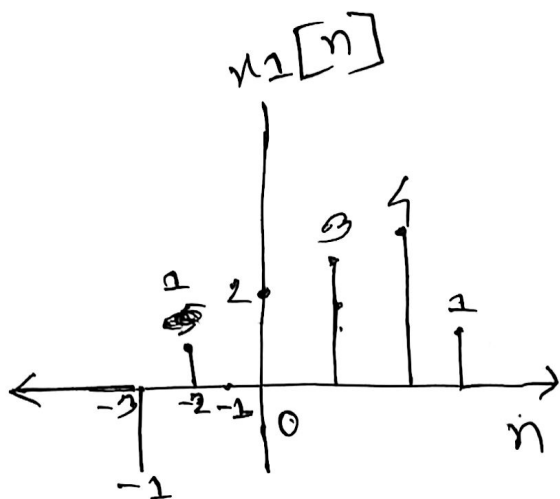
$$\boxed{a > 1} \quad / \quad \boxed{0 < a < 1} \quad / \quad a (|a| < 0) :$$

1.5 Signal Operations on DT signal

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1. Addition: Signal Addition is performed sample-by-sample (point-to-point) at identical time indices n . Consider the signals $x_1[n]$ and $x_2[n]$ and the addition of these signals at every n is represented as $x[n] = x_1[n] + x_2[n]$



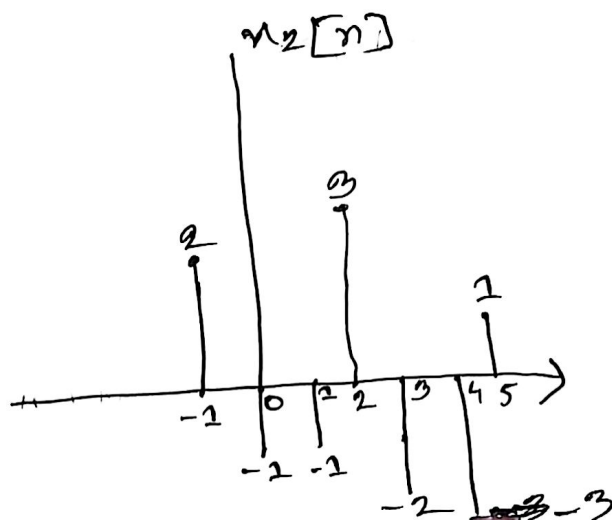
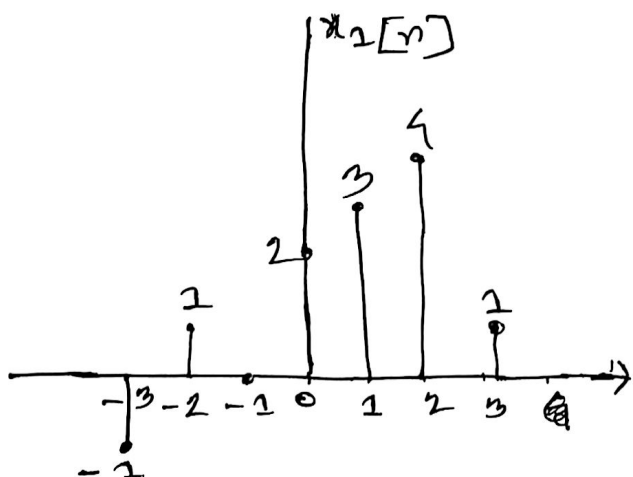


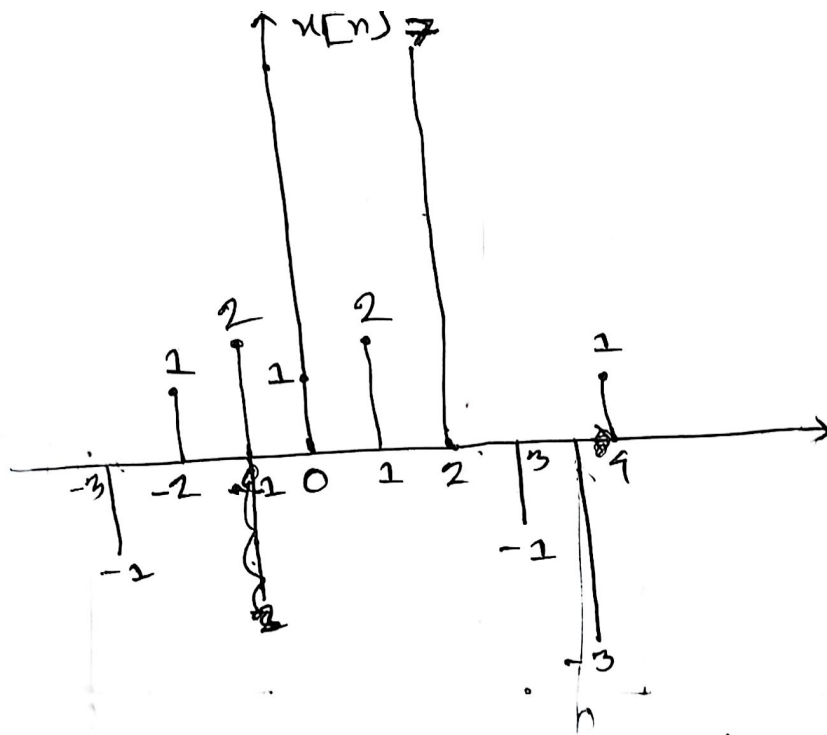
if,

$$x_1[n] = \{-1, 2, 0, 2, 3, 4, 1\}$$

$$x_2[n] = \{2, -1, -1, 3, -2, -3, 1\}$$

$$x[n] = \{-1, 2, 2, 2, 2, 7, -1, -3, 1\}$$





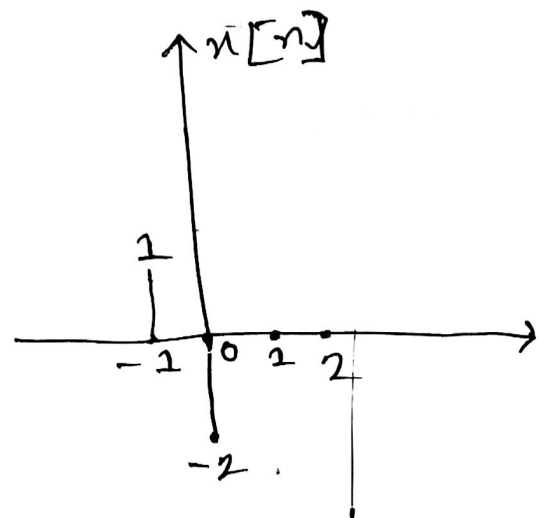
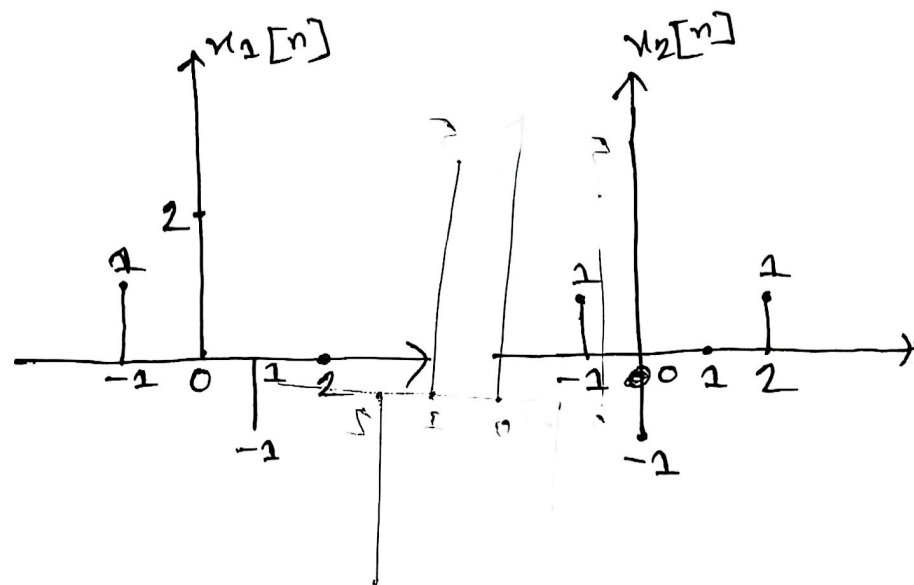
2. Multiplying: Signal Multiplication is also known as an element-wise operation carried out at matching time indices n . Mathematically, $x[n] = x_1[n] \cdot x_2[n]$

$$x_1[n] = \{1, 2, -1, 0\}$$

$$x_2[n] = \{1, -1, 0, 1\}$$

$$x[n] = x_1[n] \cdot x_2[n]$$

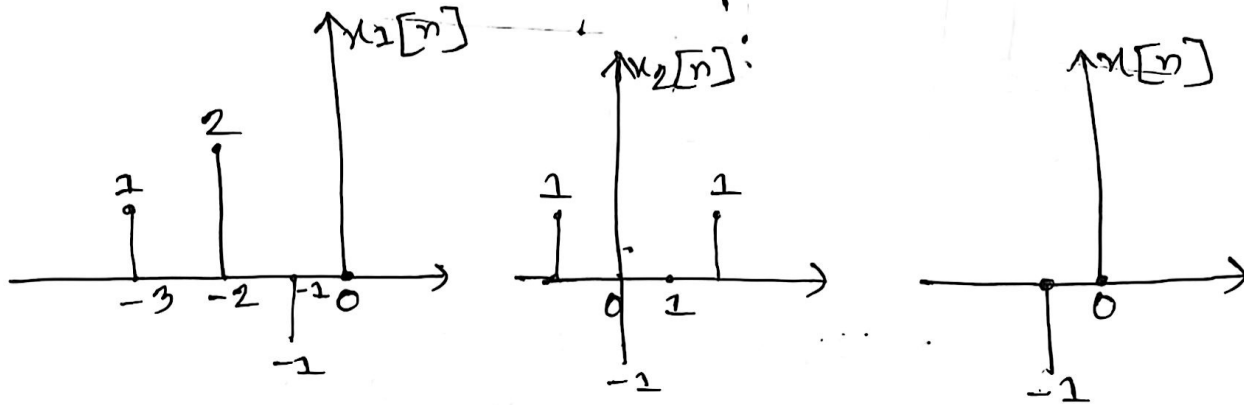
$$= \{1, -2, 0, 0\}$$



$$x_1[n] = \{1, 2, -1, 0\}$$

$$x_2[n] = \{1, -1, 0, 1\}$$

$$x[n] = x_1[n] \cdot x_2[n] = \{-1, 0\}$$



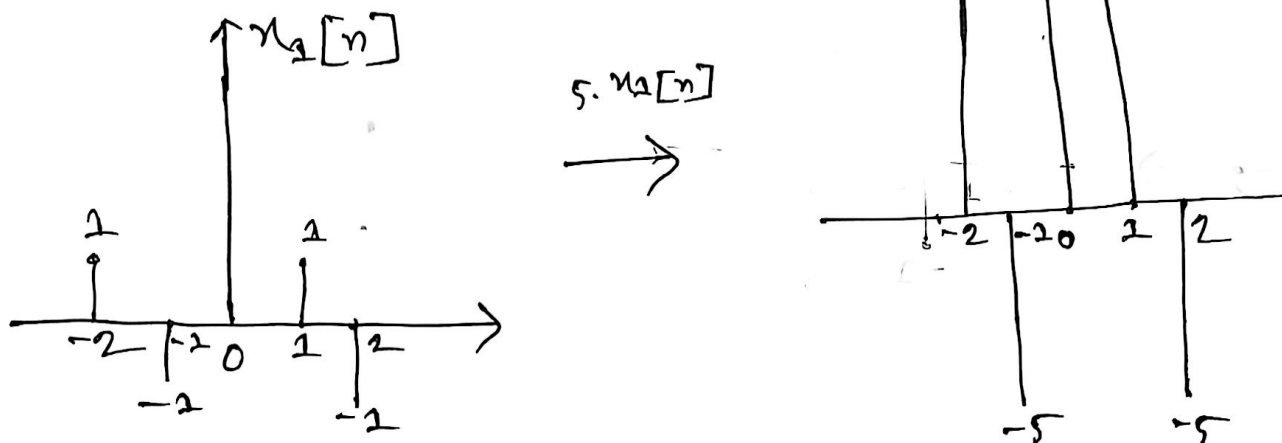
3. Amplitude Scaling: Amplitude scaling involves multiplying every sample of a single sequence by a constant scalar factor A . Mathematically:

$$x_2[n] = A \cdot x_1[n]$$

same value Gain 2/5

$$x_1[n] = [1, -1, 0, 1, -1]$$

$$x_2[n] = 5 \cdot x_1[n]$$

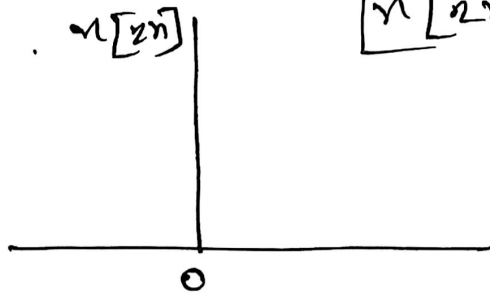
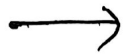
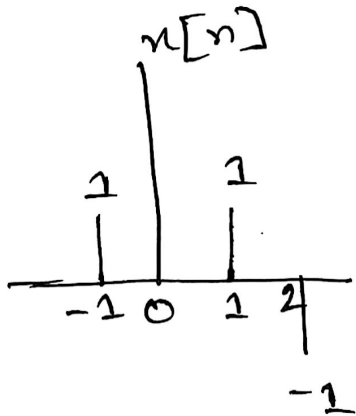


4. Time Scaling: Time Scaling modifies the time index n itself, changing the speed or density of the signal.

$$x[n] = \{1, 0, 1, -1\} \longrightarrow \text{formulas:}$$

$$\text{even } [x[2n]]$$

$$\text{odd } \begin{cases} x[2n+1] & \text{if } n < 0 \\ x[2n-1] & \text{if } n > 0 \end{cases}$$



$$x[2n] =$$